



STANDARD OPERATING PROCEDURE (SOP)

Project Management, Implementation & Assessment Framework for NAAC

Submitted By

Field	Details
Name	_____
Department	_____
Designation	_____
Institution	_____
Academic Year	_____

1. INTRODUCTION

This Standard Operating Procedure (SOP) establishes a structured framework for teaching and implementing Project Management through practical projects, assessment mechanisms, documentation practices, and industry-standard tools.

2. OBJECTIVES

The objectives of this SOP are:

- To standardize project execution procedures
- To teach Project Management through practical implementation
- To align projects with syllabus outcomes and COs
- To establish structured assessment mechanisms
- To improve documentation and evidence collection
- To ensure timely completion of project milestones

3. SAMPLE PROJECT USED FOR IMPLEMENTATION

Project Title

“Student Attendance Management System”

The project was used as a practical example to teach students complete project lifecycle management including planning, execution, monitoring, testing, documentation, and assessment.

4. PROJECT MANAGEMENT FRAMEWORK

4.1 Project Phases

Phase	Activity
Phase 1	Problem Identification
Phase 2	Literature Survey
Phase 3	Planning & Design
Phase 4	Development / Implementation
Phase 5	Testing & Validation
Phase 6	Documentation & Presentation

5. PHASE-WISE IMPLEMENTATION

Phase	Main Activities	GitHub	Jira Software	Google Drive	Assessment Weightage	SYLLABUS Mapping
Phase 1 – Problem Identification	Topic selection, problem statement, objective definition	Git ,Github overview Create repository, add README	Create project board, create initial tasks/issues	Store topic approval documents	10%	UNIT-1
Phase 2 – Literature Survey	Research papers, existing system analysis	Upload reference links/documentation	Assign research tasks and deadlines	Store PDFs, review notes, survey reports	15%	UNIT 1,2
Phase 3 – Planning & Design	Requirement analysis, UML diagrams, architecture design	Upload design files, diagrams	Sprint planning, task allocation	Store flowcharts, design documents	20%	UNIT-3
Phase 4 – Development / Implementation	Coding, module development, integration	Source code management, version control, branches & commits	Track development progress, bug tracking	Store screenshots, progress reports	30%	UNIT-3
Phase 5 – Testing & Validation	Unit testing, debugging, validation	Upload tested code and release versions	Create testing tickets, bug reports	Store test cases and testing reports	15%	UNIT-3
Phase 6 – Documentation	Final report, PPT,	Final project repository with	Close all tasks and generate	Store final documentation	10%	UNIT-3

Phase	Main Activities	GitHub	Jira Software	Google Drive	Assessment Weightage	SYLLABUS Mapping
on & Presentation	viva preparation	complete code	progress reports	on, PPTs, viva files		

Assessment Area	Weightage
Problem Identification & Research	25%
Planning & Design	20%
Development & Technical Contribution	30%
Testing & Validation	15%
Documentation & Presentation	10%

6. TEAM FORMATION & ROLE ASSIGNMENT

Team Structure

Role	Responsibility
Team Leader	Coordination and planning
Developer	Technical implementation
Documentation Lead	Reports and records
Testing Member	Validation and testing

Team Size

- Minimum: 3 Members
- Maximum: 5 Members

7. PROJECT MANAGEMENT TOOLS USED

GitHub

Used for:

- Source code management
- Version control
- Repository maintenance
- Tracking daily code commits

Activities Performed

- Repository creation
- Code commits & pushes
- Branch management
- Source code uploads

Jira Software

Used for:

- Task allocation
- Sprint planning
- Progress tracking
- Bug reporting

Example Task Allocation

Task	Assigned To
Login Module	Student A
Database Design	Student B
Testing	Student C

Task statuses monitored:

- To Do
- In Progress

- Completed

Google Drive

Used for:

- Documentation storage
- File sharing
- Backup maintenance
- Review record management

Files Maintained

- Synopsis
- Literature Review
- Weekly Reports
- PPTs
- Final Documentation

8. SYLLABUS & COURSE OUTCOME (CO) MAPPING

Activity	CO1	CO2	CO3	CO4
Requirement Analysis	✓			
Design Process		✓	✓	
Development			✓	✓
Testing & Validation		✓	✓	✓
Documentation	✓			✓

10. SUBMISSION SCHEDULE & MILESTONES

Week Deliverable

Week 1 Topic Approval

Week 2 Synopsis Submission

Week 4 Literature Review

Week 6 Design Submission

Week 8 Mid Review

Week 10 Prototype Demonstration

Week 12 Final Report Submission

Week 13 Presentation & Viva

11. DELIVERABLES

The following deliverables were maintained:

- Synopsis
- Literature Review
- Design Diagrams
- Source Code
- Progress Reports
- Testing Reports
- Presentation Slides
- Final Project Report
- GitHub Repository Links

12. DOCUMENTATION REQUIREMENTS

Required Documents

- Attendance Records
- Progress Review Sheets

- Evaluation Reports
- Meeting Minutes
- Weekly Status Reports
- Code Printouts
- Final Project Report

File Naming Format

Department_Project_Date

Example:

CSE_Project_2026

13. ASSESSMENT & EVALUATION FRAMEWORK

13.1 Assessment Objectives

The assessment process was designed to:

- Evaluate technical knowledge and implementation skills
- Measure innovation and problem-solving capability
- Assess teamwork and communication skills
- Monitor milestone completion and project quality

13.2 Assessment Components

Assessment Type	Weightage
Continuous Assessment	20%
Mid-Term Review	20%
Final Project Evaluation	30%
Documentation Assessment	15%
Presentation & Viva	15%

13.3 Continuous Assessment Parameters

Parameter	Evaluation Criteria
Attendance & Participation	Regular involvement in project activities
Task Completion	Timely completion of assigned tasks
Technical Contribution	Coding and implementation quality
Team Coordination	Collaboration and communication
Documentation Updates	Maintenance of reports and records

13.4 Project Evaluation Rubric

Criteria	Weightage
Innovation	20%
Technical Quality	25%
Documentation	15%
Team Coordination	10%
Presentation Skills	10%
Testing & Validation	20%

13.5 Tool-Based Assessment

Tool	Assessment Activity
GitHub	Code contribution, commit history, repository management
Jira Software	Task completion, sprint tracking, issue management
Google Drive	Documentation quality, report submission, record maintenance

13.6 Assessment Methods

Direct Assessment

- Internal reviews
- Coding evaluation
- Demonstration of working model
- Viva voce examination
- Final project presentation

Indirect Assessment

- Student feedback
- Peer assessment
- Industry expert suggestions
- Faculty observations

13.7 Outcome-Based Assessment Mapping

Activity	CO1	CO2	CO3	CO4
Requirement Analysis	✓			
Design & Planning		✓	✓	
Development			✓	✓
Testing & Validation		✓	✓	✓
Documentation & Presentation	✓			✓

14. REVIEW MECHANISM

Review Committee

- Head of Department
- Project Coordinator
- Faculty Guide
- External Expert (if required)

Review Frequency

- Weekly Monitoring
- Monthly Evaluation
- Final Review

15. PROJECT MANAGEMENT FLOW

Problem Identification

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Research & Survey

↓

Planning & Design

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Development Using GitHub

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Task Tracking Using Jira

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Testing & Validation

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Documentation in Google Drive

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Assessment & Final Presentation

16. FUTURE SCOPE & INTERDISCIPLINARY INTEGRATION

Projects should include:

- Future enhancement possibilities
- Industry applications

- Research opportunities
- Interdisciplinary integration
- Startup or patent scope

17. OUTCOMES ACHIEVED

Implementation of this framework resulted in:

- Improved practical learning
- Better teamwork and coordination
- Enhanced technical skills
- Strong documentation practices
- Continuous assessment evidence generation
- Effective NAAC compliance support

18. CONCLUSION

This SOP establishes a structured methodology for teaching Project Management through practical implementation, assessment, documentation, and industry-standard tools. The framework supports experiential learning, quality enhancement, systematic evaluation, and continuous academic improvement aligned with NAAC requirements.

19. SIGNATURES

Prepared By Reviewed By Approved By

Signature: _____

Date: _____